

EasyWGS: a reproducible cross-platform workflow for bacterial isolate whole-genome sequencing

1 Installation & setup

Software & database installation (module 00)

Conda / mamba environments

- easywgs (main), longread, checkm2, gunc, bakta, eggnog

Acquisition & read QC

- fastp, FastQC, MultiQC
- porechop, chopper, Filtlong, NanoPlot
- NCBI datasets, SRA toolkit

Two-layer decontamination

- CLEAN, Kraken2/Bracken, BBDuk
- CheckM2, GUNC, FCS-GX, BlobToolKit
- k2\_standard & FCS-GX databases

Assembly & polishing

- Unicycler, SPAdes, Flye, Canu, dragonflye
- Tricycler, minimap2+Racon, Medaka, Pilon
- circlator, QUAST, seqkit, assembly-stats

Annotation & typing

- Prokka, Bakta, Prodigal, eggNOG-mapper
- mlst; chewBBACA (cgMLST)
- Kleborate/Kaptive, ECTyper, ShigEiFinder
- SeqSero2, SISTR
- PubMLST & species cgMLST schemas

AMR / virulence / MGE

- abricate (ResFinder/VFDB/CARD/MEGARes)
- AMRFinderPlus, RGI, PointFinder
- mob-suite, PlasmidFinder, IntegronFinder
- ISEScan, mobileOG, genomad, PhiSpy, CRISPRCasFinder
- PLSDB, ResFinder, VFDB, CARD

Comparison & evolution

- Panaroo, Roary; mash, cd-hit
- snippy, Gubbins, snp-sites, snp-dists
- IQ-TREE 3, FastTree, MAFFT, MUMmer
- TreeTime (clock, ancestors, murgation)

Mapping & genomic GWAS

- BWA, minimap2, samtools/bcftools, htslib
- Qualimap, vcf2phylip, tabix
- Scoary, PLINK, pyseer; R (BH/Bonferroni)

Visualization

- GrapeTree; iTOL datasets; ggtree
- ComplexHeatmap; ggplot2

Workflow drivers

run\_assembly.sh / run\_mapping.sh  
run\_all.sh chains every stage; 99\_report merges results

Legend

- software
- database
- data flow

2 QC & two-layer decontamination

Data analysis pipeline (modules 01–13, 99)

Raw reads + metadata (config/samplesheet.csv): Illumina paired-end | Oxford Nanopore | PacBio | hybrid

config/traits.csv

per-isolate id, platform, species, R1/R2, long reads, reference, date, country, phenotype

01 Quality control

Short reads: fastp adapter/quality trimming and report Long reads: porechop/chopper, Filtlong, NanoPlot

trimmed FASTQ + FastQC/MultiQC

02 Read-level decontamination (layer 1: keep-by-target)

CLEAN retains only target-taxon reads; Kraken2/Bracken scouts composition; BBDuk removes reference/adaptor hits

clean, single-source reads (FASTQ)

ROUTE A — assembly-based (modules 03–09)

ROUTE B — reference mapping (module 12)

03 Assemble & polish

Unicycler/SPAdes for short & hybrid; Flye/Canu/Tricycler for long; minimap2+Racon, Medaka, Pilon polish; circlator circularizes

contigs FASTA

04 Assembly QC & decontam (layer 2)

QUAST/seqkit stats; CheckM2 completeness/contamination; GUNC chimerism; FCS-GX removes foreign contigs/fragments

clean genome FASTA

05 Structural & functional annotation

Prokka/Bakta/Prodigal genes; eggNOG-mapper GO/KEGG/COG

GFF, GenBank, proteins

06 Typing: MLST, serotype, cgMLST

mlst seven-gene ST; Kleborate/Kaptive (Klebsiella K/O), ECTyper/ShigEiFinder (E. coli–Shigella), SeqSero2/SISTR (Salmonella); chewBBACA cgMLST alleles

ST / serotype / allele profiles

07 AMR, virulence & mobile elements

abricate/AMRFinderPlus/RGI/PointFinder; plasmid (mob-suite, PlasmidFinder/PLSDB); integron, IS, ICE, phage/GI, CRISPR

resistance & MGE tables

08 Pangenome

Panaroo (Roary): core/accessory partition and graph

gene presence–absence matrix

09A Core SNPs from assemblies

snippy/Gubbins recombinant masking, snp-sites, snp-dists

core alignment, SNP distance

12.1 Map reads to a common reference

BWA-MEM for Illumina/hybrid, minimap2 -ax map-on/map-pb for long reads; samtools sort/index, flagstat, depth -a; coverage breadth/depth; Qualimap BAM QC

sorted+indexed BAM, coverage summary

12.2 Joint variant calling & filtering

bcftools mpileup (-q20 -Q20) | call -mv | norm | filter QUAL/DP; keep bi-allelic SNPs (missingness, MAC); bgzip+tabix

raw/filtered/bi-allelic VCF

SNP matrix & alignment-free check

vcf2phylip core alignment; snp-dists pairwise matrix; mash for rapid neighbors

genotype matrix / SNP distances

When to choose mapping

Clonal outbreak tracing and large surveillance sets with a high-quality close reference; one common coordinate system, sensitive SNP/indel calls at moderate depth, no assembly step.

Route A is preferred for accessory genes, plasmids, MGE and broad lineage diversity

Comparative genomics & evolution (modules 09–10)

Core/accessory calls, cgMLST alleles and core-SNP alignments/distances converge into one phylogeny:

IQ-TREE 3 (FastTree for quick look) → 10 TreeTime: molecular clock, dated time tree, ancestral sequence reconstruction and trait murgation (country/host/phenotype)

13 Microbial genomic GWAS & post-GWAS (input: gene presence–absence from 08, VCF from 12, traits.csv)

13.1 Scoary

gene-level pan-GWAS, Fisher + permutation, pairwise correction

13.2 PLINK

SNP logistic/linear model with IBS/MDS population covariates

13.3 pyseer

mixed model with SNP/mash distance kernel; gene, SNP, optional k-mer

13.4 post-GWAS in R: BH/Bonferroni correction, QQ & Manhattan plots, merged hit table

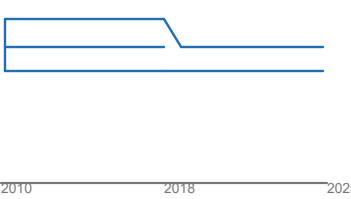
4 Integration, visualization & reporting

Statistics, visualization & reporting (11, 99)

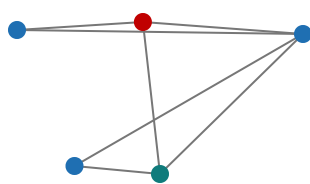
Phylogeny (ggtree, IQ-TREE 3)



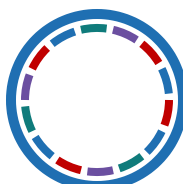
Time-scaled tree (TreeTime)



GrapeTree / MST clusters



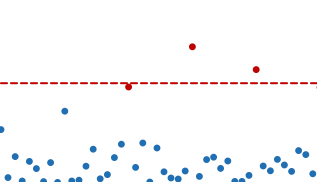
iTOL annotated rings



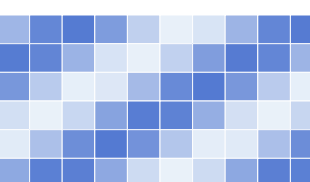
ggtree annotation panels



GWAS QQ & Manhattan (13.4)



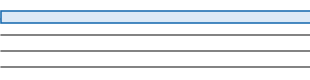
AMR / allele heatmaps



Metadata maps & timeline



99 merged, typed report



Integrated, reproducible outputs (99\_report)

- Clean single-source reads and assemblies (FASTQ/FASTA)
- ST, species serotype and cgMLST allele profiles
- AMR, virulence, plasmid and MGE annotations
- Pangenome, core-SNP and cgMLST distance matrices
- Sorted/indexed BAM, coverage metrics and filtered VCF
- IQ-TREE 3 phylogeny and TreeTime-dated time tree
- GWAS hits with BH/Bonferroni, QQ and Manhattan plots
- GrapeTree/iTOL/ggtree bundles and merged report